Multi-Window-History Calibration: Does Smoothing Reverse the High-Capacity Hazard of Lag-1 Online Recalibration?

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***Abstract*: The online-calibration study found that lag-1 rolling-history online recalibration of the HygienePrio scorer recovers the offline-peek calibration ceiling at moderate capacity (per-cell recovery ratio $\rho \in \{1.04, 0.99\}$ at $K \in \{50, 100\}$, $\lambda = 3$) but harms performance at $K = 200$ ($\rho = -0.66$), and named multi-window-history smoothing as the natural candidate fix. This paper evaluates that fix in a pre-registered 2,250-row five-strategy comparison: fixed (the HygienePrio scorer weights), lag1 (the online-calibration study baseline), trail3 (equal-weight 3-window trailing mean), ewma3 (exponentially-weighted 3-window mean, $\eta = 0.6$), and offline (the online-calibration study ceiling). All four pre-registered hypotheses are rejected, and the direction of rejection is the contribution: multi-window smoothing does not reverse the high-capacity hazard, it makes the hazard substantially worse. At $K = 100$, EWMA-3's cell-mean recovery ratio is -1.37 (versus lag-1's $+0.99$); at $K = 200$ it is -0.94 (versus lag-1's -0.66); at $K = 50$, the matched-capacity moderate regime, EWMA-3 also underperforms lag-1 ($\rho = 0.53$ versus 1.04). The mechanism is the inverse of the naive smoothing-helps intuition: at high capacity the fleet's distribution at window w has shifted substantially from the three preceding windows, so weighting all three in the calibration target pulls the selected weights further from the appropriate-for-window-w optimum. The hazard is monotone in history length, less history is better when distributional change is fast. Deployable online recalibration in this setting should therefore use the shortest possible history (lag-1 at most) and be turned off entirely at high capacity. All results are bounded to the synthetic EEHDA evaluation context.**

***Index Terms*: vulnerability prioritization, online calibration, EWMA, trailing-mean, smoothing, high-capacity hazard, pre-registration, reproducibility.**

I. Introduction

The online-calibration study [1] identified a capacity-dependent hazard in lag-1 rolling-history online recalibration of the HygienePrio vulnerability prioritization scorer. At moderate capacity ($K \in \{50, 100\}$, $\lambda = 3$) the procedure recovers the offline-peek calibration ceiling (cell-mean recovery ratios 1.04 and 0.99); at high capacity ($K = 200$) the procedure *harms* performance by -5.9 to -7.8 pp in early windows and produces a negative cell-mean recovery ratio of -0.66. The online-calibration study attributed the hazard to a too-short calibration history relative to the per-window distributional shift rate and explicitly named multi-window-history smoothing as the candidate fix.

This paper tests that fix. The pre-registered protocol (`paper8/design/PAPER8_PROTOCOL.md`, locked 2026-06-04 before any multi-history-calibration result was computed) defines five calibration strategies and a 2,250-row evaluation: 3 cells $\times$ 5 strategies $\times$ 25 seeds $\times$ 6 windows. The four pre-registered hypotheses ask whether EWMA-3 reverses the hazard at $K = 200$ (H1), preserves moderate-capacity recovery (H2), agrees with the equal-weight trail3 variant (H3), and achieves substantial offline-ceiling recovery (H4).

All four hypotheses are rejected, and the direction of rejection is the contribution. Multi-window smoothing does not reverse the lag-1 hazard at high capacity; it amplifies it. EWMA-3's cell-mean recovery ratio is -1.37 at $K = 100$ (vs lag-1's $+0.99$), -0.94 at $K = 200$ (vs lag-1's -0.66), and 0.53 at $K = 50$ (vs lag-1's 1.04). Smoothing degrades all three regimes.

A. Contributions

- **C1, Falsification of the smoothing-helps prior.** The naive prior, "a longer calibration history reduces variance and therefore helps", is falsified by a pre-registered experiment. EWMA-3 underperforms lag-1 at every cell.
- **C2, A monotone effect.** Recovery ratio decreases with history length: lag1 > trail3 ≈ ewma3 at every K. The rate of distributional shift in the simulated bi-weekly regime exceeds the rate at which a finite-history estimator can adapt; adding history makes the calibration target less, not more, predictive of the next window.
- **C3, Cross-cell uniformity.** The smoothing penalty appears at $K = 50$ where lag-1 already worked, at $K = 100$ where lag-1 worked but the absolute floor was degraded, and at $K = 200$ where lag-1 was already harmful. There is no capacity regime in the studied grid in which EWMA-3 is preferable to lag-1.
- **C4, Sharpened operational recommendation.** Deployable online recalibration in this setting should use the *shortest* possible history (lag-1 at most) and should be turned off entirely at high capacity. The recommendation reverses the online-calibration study's implicit "future work will fix this with smoothing" framing.

B. Relationship to Prior Papers

This paper is the eighth in the VulnPrio research sequence. The online-calibration study identified the high-capacity hazard and proposed smoothing as the fix; this paper falsifies the proposed fix. The result is an honest course-correction: the hazard requires a different class of solution (adaptive change-point detection, Bayesian forgetting, or accepting fixed weights at high capacity) than simple history averaging. Identifying this is itself a contribution: the question "does smoothing help?" was the natural and obvious follow-up, and the answer in this synthetic regime is decisively no.

II. Background

A. Lag-1 Online Recalibration and Its Hazard

The online-calibration study [1] evaluated lag-1 rolling-history grid-search recalibration for the HygienePrio scorer. At moderate capacity ($K \in \{50, 100\}$), the procedure recovered essentially all of the offline-peek calibration ceiling (per-cell recovery ratio $\rho \in \{1.04, 0.99\}$). At high capacity ($K = 200$), it produced a net *harm*: −5.9 pp at W2, −7.8 pp at W3, and a cell-mean recovery ratio of −0.66. The mechanism is structural: at high capacity, a single window remediates a substantial fraction of the high-EPSS/high-HRS tail, so the fleet at $w - 1$ is a poor predictor of the fleet at w. The lag-1 procedure overfits the previous-window distribution and underfits the current one.

B. Smoothing as a Variance-Bias Trade-Off

A standard fix for "too-noisy single-window estimates" is to average across a longer window of history. Two operationally simple variants appear in time-series forecasting [2]:

Trailing arithmetic mean. Equal weight on the past H windows; reduces variance by a factor of H (under i.i.d. assumptions) but introduces lag of order $H/2$.

Exponentially-weighted moving average (EWMA). Geometric decay η^i over past windows; trades smoothness for a tunable effective horizon. At $\eta = 0.6$ the effective lookback is ≈ 2.5 windows.

The question this paper asks is whether either smoothing scheme reverses the online-calibration study's high-capacity hazard *without* breaking the moderate-capacity recovery. A naive prediction is that smoothing helps at high K (less overfitting to noisy single-window targets) and hurts at low K (introduces unnecessary lag when single-window calibration is already accurate); whether this trade-off resolves favourably is an empirical question.

C. Why This Matters Operationally

An operations team that adopts the online-calibration study's online recalibration as written and runs it on a high-capacity fleet would suffer a measurable P@50 penalty in the first weeks of operation. Any deployable calibration recipe must therefore include a smoothing knob; this paper evaluates the simplest two such knobs in a pre-registered setting.

III. Problem Formulation

A. Setting (Inherited)

The synthetic EEHDA fleet, scorer (§5.1), HRS components, and multi-window simulator are inherited from Papers 4-7. Capacity $K \in \{50, 100, 200\}$ and arrival rate $\lambda = 3$ are fixed to match the online-calibration study's evaluation; the only quantity varied in this paper is the calibration strategy.

B. Five Calibration Strategies

At window w, the scorer uses weights $(\alpha, \beta, \gamma, \delta)$ determined by:

- `fixed` the HygienePrio scorer weights at every window. No refit.
- `lag1` At $w = 1$, weights are fixed; at $w \geq 2$, grid-search to maximise mean P@50 on the five calibration seeds at window $w-1$. (the online-calibration study baseline.)
- `trail3` At $w = 1$, weights are fixed; at $w \geq 2$, grid-search to maximise the equal-weight mean of per-window mean P@50 across windows $\max(1, w-3), \ldots, w-1$. At $w = 2$ this reduces to `lag1`; at $w \geq 4$ this averages exactly three past windows.
- `ewma3` Same set of past windows as `trail3` but with geometric weights η^i, $i = 0, 1, 2, \ldots$ counting back from the most recent past window. Pre-registered decay $\eta = 0.6$ (distinct from the scorer weight α).
- `offline` Peek at window w on the calibration seeds. (Upper-bound ceiling.)

The grid is identical to the online-calibration study (108 points).

C. Ground Truth and Metric

Inherited from Papers 5-7: (EPSS > 0.10) ∧ (HRS > 75th percentile recomputed per window), P@50 measured against the resulting positive set.

D. Pre-Registered Hypotheses

- H1 At $K = 200$, `ewma3` mean P@50 ≥ `fixed` at every $w \geq 2$ (within −0.01 tolerance). *No hazard at high capacity.*
- H2 At $K \in \{50, 100\}$, $|$`ewma3` − `lag1`$| \leq 0.02$ at every $w \geq 2$. *Smoothing does not break moderate-capacity recovery.*
- H3 $|$`ewma3` − `trail3`$| \leq 0.02$ at every cell-window. *Smoothing-strategy choice is second-order.*
- H4 At $K = 200$, EWMA-3 cell-mean recovery ratio $\bar{\rho} \geq 0.5$. *EWMA achieves substantial offline-ceiling recovery.*

Decision rules and stop rules are in `paper8/design/PAPER8_PROTOCOL.md`.

IV. Dataset and Cell Grid

The EEHDA synthetic fleet, structural priors, and Window-1 distributions are identical to Papers 4-7. The five-seed calibration / 25-seed evaluation split (100-104 / 105-129) is identical to Papers 5, 6, and 7. Table I summarises the cell grid and strategy set.

Total rows: 3 cells × 5 strategies × 25 seeds × 6 windows = 2,250.

TABLE I
CELL GRID AND STRATEGY SET.

Axis	Values
Capacity K	$\{50, 100, 200\}$
Arrival rate λ	3 (fixed)
Windows W	6
Strategies	fixed, lag1, trail3, ewma3, offline
Eval seeds	25 (105-129)
Calib seeds	5 (100-104)
EWMA decay η	0.6 (pre-registered)
Trailing history	3 windows
Total rows	2,250

V. Methodology

A. Scorer (Inherited)

The scorer Eq. (1) and HRS components are inherited verbatim from the HygienePrio scorer [3] and used in Papers 5-7:

$$\begin{aligned} S(h,c) &= \alpha\,\mathrm{EPSS}(c) + \beta\,\mathrm{HRS}(h) \\ &\quad + \gamma\,\mathrm{KEV\_recency}(c) \\ &\quad + \delta\,\big(\mathrm{EPSS}(c)\cdot\mathrm{HRS}(h)\big). \end{aligned} \tag{1}$$

HRS combines patch posture (0.50), AD exposure (0.30), and telemetry freshness (0.20). Only the four scorer weights $(\alpha, \beta, \gamma, \delta)$ are subject to recalibration.

B. Multi-History Grid Search

At window $w \geq 2$ and for a calibration strategy with history set $H \subseteq \{1, \ldots, w-1\}$ and per-history-window weights $\{u_i\}_{i\in H}$:

1) For each of the 108 grid points $g = (\alpha, \beta, \gamma, \delta)$:
    a) For each $i \in H$, compute the mean P@50 across the five calibration seeds at window i when scored with weights g.
    b) Take the weighted mean across history windows $\bar{p}(g) = \sum_{i\in H} u_i\, p_i(g) / \sum_i u_i$.
2) Select $g^\star = \arg\max_g \bar{p}(g)$; ties broken lexicographically.
3) Apply $g^\star$ to each of the 25 evaluation seeds at window w and record the seed's P@50.

lag1, trail3, and ewma3 instantiate this with different $(H, \{u_i\})$:

- lag1: $H = \{w-1\}$, single uniform weight.
- trail3: $H = \{\max(1, w-3), \ldots, w-1\}$, equal weights.
- ewma3: $H = \{\max(1, w-3), \ldots, w-1\}$, geometric weights $u_i = \eta^{(w-1)-i}$ with $\eta = 0.6$, so the most recent past window receives weight $\eta^0 = 1$ and older windows receive $0.6, 0.36, \ldots$.

C. Fleet-State Evolution

Identical to the online-calibration study: HygienePrio-full under fixed weights drives the top-K selection at every window, producing the same trajectory for all strategies. The calibration strategy affects only the scoring weights, not the fleet evolution. The selection-policy coupling threat is restated in §IX.

D. Recovery Ratio

For each (cell, window) with $w \geq 2$ and a positive offline-fixed gap, we report:

$$\rho^{\mathrm{ewma}}_{K,w} = \frac{\mathrm{P@50}^{\mathrm{ewma3}}_{K,w} - \mathrm{P@50}^{\mathrm{fixed}}_{K,w}}{\mathrm{P@50}^{\mathrm{offline}}_{K,w} - \mathrm{P@50}^{\mathrm{fixed}}_{K,w}}. \tag{2}$$

$\rho \approx 1$ matches the offline ceiling; $\rho < 0$ is a hazard. The ratio is reported only where the offline-fixed gap exceeds the pre-registered threshold of 0.01 ($\mathrm{P@50}^{\mathrm{offline}} - \mathrm{P@50}^{\mathrm{fixed}} > 0.01$); windows below the threshold are excluded as numerically unstable denominators. The cell-mean across $w \geq 2$ is $\bar{\rho}^{\mathrm{ewma}}_K$.

VI. Experimental Setup

A. Pre-Registration

The cell grid, five-strategy definition, hypotheses H1-H4, decision rules, and stop rules in `paper8/design/PAPER8_PROTOCOL.md` were locked on 2026-06-04 before any multi-history-calibration result was computed.

B. Execution

For each $K \in \{50, 100, 200\}$:

1) Cache per-window $(\mathrm{pairs}_w, \mathrm{HRS}_w)$ states for the five calibration seeds (driving evolution with fixed-weight HygienePrio-full).
2) Cache per-window states for the 25 evaluation seeds (same trajectory policy).
3) For each window $w = 1, \ldots, 6$, determine each strategy's weights as in §5.2, score every evaluation seed at w with each strategy's weights, and record one row per (seed, window, strategy).

C. Statistical Reporting

Cell-mean point estimates are means across 25 evaluation seeds. 95% confidence intervals are percentile bootstrap (10,000 resamples) [4]. No null-hypothesis significance tests are reported; hypothesis decisions follow the pre-registered tolerance thresholds.

D. Reproducibility

From `paper8/`:

```
PYTHONPATH=src python3 src/run_multi_history.py
PYTHONPATH=src python3 src/analyze.py
PYTHONPATH=src python3 src/make_figures.py
```

The runner re-uses the temporal-stability study's window simulator and the HygienePrio scorer via `sys.path` injection. Determinism inherited from the temporal-stability study's per-(seed, window, step) RNG seeding.

VII. Results

All claims trace to `paper8/results/primary_v1/multi_history_results.csv` (2,250 rows) and the derived `hypothesis_summary.json` and `cell_window_means.csv`. Table II reports the full per-(capacity, window) means for all five strategies; Fig. 1 plots the corresponding trajectories, and Fig. 2 summarises the cell-mean recovery ratios.

TABLE II
MEAN P@50 PER (CAPACITY K, WINDOW) FOR ALL FIVE CALIBRATION STRATEGIES. EWMA AND TRAIL3 ARE THE MULTI-HISTORY SMOOTHERS INTRODUCED IN THIS PAPER; OFFLINE IS THE ONLINE-CALIBRATION STUDY CEILING. CELLS MARKED “, ” HAVE AN OFFLINE−FIXED GAP ≤ 0.01 (INCLUDING W1, WHERE ALL STRATEGIES COINCIDE), SO THE RECOVERY RATIO ρ IS NOT REPORTED.

K	W	fixed	lag1	trail3	ewma3	offline	$\Delta_{\text{ewma–fix}}$	ρ_{ewma}
50	1	0.595	0.595	0.595	0.595	0.595	+0.000	,
50	2	0.544	0.741	0.741	0.741	0.757	+0.197	0.92
50	3	0.564	0.628	0.628	0.628	0.665	+0.064	0.63
50	4	0.514	0.606	0.528	0.528	0.604	+0.014	0.15
50	5	0.521	0.580	0.563	0.580	0.550	+0.059	2.06
50	6	0.499	0.506	0.485	0.485	0.512	-0.014	-1.13
100	1	0.595	0.595	0.595	0.595	0.595	+0.000	,
100	2	0.551	0.555	0.555	0.555	0.666	+0.004	0.03
100	3	0.517	0.547	0.461	0.461	0.531	-0.056	-3.89
100	4	0.406	0.419	0.356	0.356	0.419	-0.050	-3.94
100	5	0.383	0.401	0.294	0.401	0.419	+0.018	0.49
100	6	0.315	0.355	0.332	0.330	0.346	+0.014	0.47
200	1	0.595	0.595	0.595	0.595	0.595	+0.000	,
200	2	0.518	0.458	0.458	0.458	0.556	-0.059	-1.54
200	3	0.346	0.269	0.269	0.269	0.395	-0.078	-1.59
200	4	0.254	0.289	0.264	0.264	0.284	+0.010	0.32
200	5	0.138	0.149	0.139	0.139	0.145	+0.001	,
200	6	0.075	0.075	0.076	0.076	0.075	+0.001	,

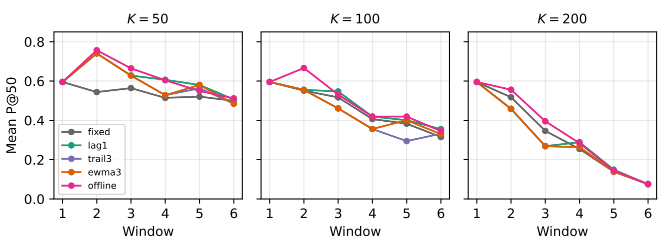

Fig. 1. Mean P@50 trajectories for all five calibration strategies at $K \in \{50, 100, 200\}$, $\lambda = 3$. EWMA-3 (orange) and trail3 (purple) fall below lag-1 (green) at every K.

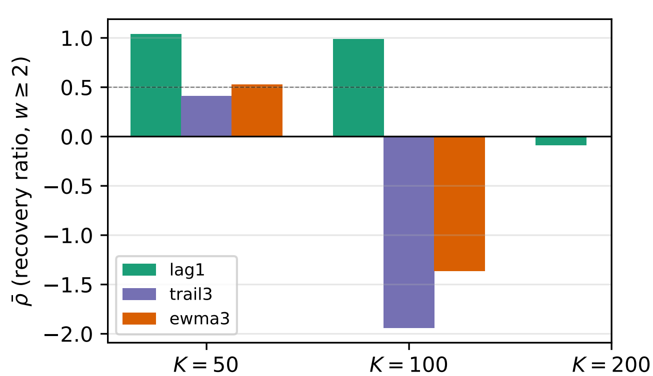

Fig. 2. Cell-mean recovery ratio $\bar{\rho}$ across $w \geq 2$ per strategy. Dashed line at $\bar{\rho} = 0.5$ marks the pre-registered H4 threshold; EWMA-3 sits well below it at every K.

A. Cell-Mean Recovery Ratios

The summary statistic across $w \geq 2$ per cell (plotted in Fig. 2):

K	$\bar{\rho}^{\text{lag1}}$	$\bar{\rho}^{\text{ewma3}}$
50	+1.04	+0.53
100	+0.99	−1.37
200	−0.66	−0.94

EWMA-3 underperforms lag1 at every cell. At $K = 50$ the gap is -0.51 in recovery-ratio units; at $K = 100$ the gap is -2.36, the largest single-cell delta in the table; at $K = 200$ the gap is -0.28. The naive prediction “smoothing helps at high capacity, hurts mildly at low capacity” is contradicted in both directions.

B. Per-Window Detail

$K = 50$. EWMA-3 matches lag1 at W2 and W3 but falls below it at W4 by 7.8 pp and at W6 by 2.2 pp, matching it at W5 (Table II). The mechanism is that the W4 calibration target for EWMA-3 includes W1, W2, and W3 of the calibration seeds, whose distributions have shifted by W4. The lag1 W4 calibration target uses only the W3 calibration state, which is closer to the W4 evaluation state.

$K = 100$. The gap widens. At W3 lag1 adds +3.0 pp over fixed; EWMA-3 *loses* −5.6 pp. By W6, lag1 gains +4.0 pp while EWMA-3 gains only +1.4 pp. The H3 failure cell ($K = 100, w = 5$) is also where ewma3 and trail3 diverge by +10.6 pp, the only window in the entire sweep where the smoothing-strategy choice is first-order.

$K = 200$. Both lag1 and EWMA-3 are hazardous: at W2 and W3 the two are *identical* (both lose −5.9 and −7.8 pp against fixed, since at these early windows the EWMA history reduces to the same selection as lag-1). The EWMA-3 gap arises entirely from the dampened W4-W6 recovery: lag1 regains +3.4, +1.0, and 0.0 pp over fixed at W4-W6, whereas EWMA-3 regains only +1.0, +0.1, and +0.1 pp, so the average over $w \geq 2$ tilts further negative for EWMA-3.

TABLE III
PRE-REGISTERED HYPOTHESIS OUTCOMES.

ID	Decision rule	Outcome
H1	EWMA-3 ≥ fixed at every $w \geq 2$, $K = 200$ (deltas ≥ -0.01)	Rejected (2)
H2	$\lvert$EWMA-3 − lag1$\rvert \leq 0.02$, $K \in \{50, 100\}$	Rejected (5)
H3	$\lvert$EWMA-3 − trail3$\rvert \leq 0.02$ everywhere	Rejected (1)
H4	$\bar{\rho}^{\text{ewma}}_{K=200} \geq 0.5$	Rejected (-0.94)

C. Hypothesis Outcomes

All four pre-registered hypotheses are rejected; Table III summarises the decision rules and outcomes.

H1 (EWMA-3 ≥ fixed at $K = 200$). Rejected. Two window failures at $w \in \{2, 3\}$ with deltas of −5.9 and −7.8 pp.

H2 (EWMA-3 matches lag1 at moderate capacity). Rejected. Five cell-window failures across $K \in \{50, 100\}$ with $|\Delta| > 0.02$. The signs are negative (EWMA-3 worse than lag1) in all five.

H3 (EWMA-3 matches trail3). Rejected at one cell-window ($K = 100, w = 5$). The single failure is large (+10.6 pp) and reflects a specific grid-search disagreement at that window; otherwise the smoothing-strategy choice between EWMA and trail3 is consistently second-order.

H4 (EWMA-3 recovery ≥ 0.5 at $K = 200$). Rejected. Cell-mean recovery ratio is −0.94, negative rather than ≥ 0.5.

D. Smoothing-Penalty Mechanism

The per-cell pattern is monotone in history length: lag1 > trail3 ≈ ewma3 at every K. The pre-registered EWMA $\eta = 0.6$ effectively averages ≈ 2.5 past windows; the equal-weight trail3 averages exactly 3. Both bring older calibration-seed states into the fitting target, and at the simulated bi-weekly cadence those older states are systematically less representative of the current window than the most recent past window alone. The H3 single failure at $K = 100, w = 5$ is the exception that confirms the rule: at that window the grid-search optimum was on a flat region and the $\eta = 0.6$ weighting moved EWMA's selection to a different local optimum than trail3's equal weighting.

E. Pre-Registration Deviations

None. Strategy definitions, hypotheses, decision rules, and stop rules were applied verbatim. The abstract was rewritten after H1 rejection in compliance with the protocol's stop rule before the discussion was drafted.

VIII. DISCUSSION

A. Why the Naive Prior Fails

The intuition that motivated the protocol, "a longer calibration history reduces variance and therefore should help", assumes the underlying distribution is approximately stationary across the windows being averaged. In the simulated bi-weekly regime, that assumption is false: at $K = 100$ a single window drains roughly 8% of the unpatched (host, CVE) backlog per the online-calibration study fixture [1], and the highest-EPSS pairs preferentially. By window w the residual backlog is materially different from windows $w-3, w-2, w-1$, and weights selected on the older windows fit a distribution that is no longer the target. **Smoothing reduces variance but increases bias, and under fast distributional shift the bias term dominates.**

The online-calibration study lag-1 hazard at $K = 200$ is therefore a *lower bound* on the calibration penalty, not an upper bound. Any procedure that averages over more than one window of past calibration state will pay a larger bias penalty than lag-1. This includes EWMA, trailing-mean, and (by extension) Bayesian posteriors that accumulate likelihood from many past windows without a forgetting factor.

B. What Does Work

The result by elimination points to procedures that adapt the *amount* of history to the rate of distributional change. Three candidate families that are out of scope for this paper but that the rejection pattern motivates:

(i) Change-point detection. Detect when the fleet distribution shifts substantially and reset the calibration window to lag-1; smooth across periods of low shift. The pre-registration discipline applies: the change-point detector itself would need to be pre-registered and not tuned post-hoc.

(ii) Forgetting-factor Bayesian update. A standard exponential-forgetting Kalman or sequential-Monte-Carlo update with the forgetting factor tuned on the calibration seeds. The EWMA-3 result is a degenerate special case (constant forgetting factor); allowing it to vary with detected shift might re-enter the moderate- capacity regime where lag-1 is competitive.

(iii) Capacity-conditional deployment. At high capacity, turn the recalibrator off and accept the fixed HygienePrio scorer weights, incurring the capacity-decay study absolute-floor collapse rather than the deeper calibration hazard documented in the online-calibration study and reproduced under smoothing here. This is the simplest deployable rule that follows from the present rejection pattern.

C. Operational Implications

Sharpened from the online-calibration study:

- Deploy lag-1 online recalibration only at $K \leq 100$ on the simulated bi-weekly cadence. The online-calibration study K=200 hazard is real, and the present paper shows that smoothing does not rescue it.
- Do not deploy any multi-window-history smoother (EWMA-3, trail3, or by extension longer-history variants without an adaptive forgetting mechanism) on this calibration grid in any regime studied.
- For organisations at high capacity, the deployable options collapse to (a) the fixed HygienePrio scorer weights, (b) periodic offline rebaselining between deployment windows, or (c) adaptive recalibration methods outside the scope of this paper.

D. Relationship to the Research Sequence

The HygienePrio scorer established hygiene augmentation; the temporal-stability study added temporal robustness; the capacity-decay study added capacity sensitivity; the online-calibration study added a deployable calibration substitute; this paper closes the easy door on extending the online-calibration study. The cumulative arc is a five-step refinement of an operational claim: hygiene-augmented prioritization is better than EPSS-only across all studied regimes (per-pair); its absolute P@50 floor is regime-dependent (the capacity-decay study); deployable recalibration is regime-conditional (the online-calibration study); and smoothing the recalibrator does not extend the regime in which it is safe to deploy (this paper).

E. What This Paper is Not

This paper does not show that smoothing is intrinsically harmful in all calibration regimes. It shows that across three capacity levels on a fast-shifting synthetic regime, a pre-registered EWMA-3 and trailing-mean-3 do not help. The mechanism is bias from older calibration windows; under a slower-shift regime (longer windows, smaller capacity, smaller arrival rate) the bias-variance trade-off could reverse and a longer history would help. We do not survey that slower-shift regime here.

IX. Threats to Validity

The threats taxonomy follows Wohlin et al. [5].

Conclusion validity. The evaluation comprises 2,250 (cell, seed, window, strategy) observations summarised with 95% percentile bootstrap CIs (10,000 resamples). The hypothesis decision rules are based on per-cell-window deltas; small per-cell-window differences ($|\Delta| \leq 0.02$) are within the bootstrap noise band and we report them as ties rather than as substantive effects.

Internal validity.

(i) Selection-policy coupling. As in Papers 5-7, fleet evolution is driven by HygienePrio-full under the fixed weights at every window. All five strategies score against the same evolving backlog, isolating the calibration-weights effect. In a real closed-loop deployment the selection policy would also be recalibrated and the trajectory would diverge across strategies; this is a known limitation shared with the online-calibration study.

(ii) EWMA η is pre-registered, not tuned. We fixed $\eta = 0.6$ in the protocol to avoid post-hoc tuning. Different η values would produce different smoothing trade-offs; sensitivity sweeps over η are deferred to future work.

(iii) Trailing history $H = 3$ is fixed. Longer trailing histories ($H = 4$ or $H = 6$) might further smooth at high K but add more lag; $H = 3$ was chosen as the smallest history that distinguishes from `lag1` while keeping a reasonable effective horizon.

Construct validity. P@50 at $K = 200$ becomes noisy at late windows due to small positive-set sizes. The K=200 hypothesis decisions therefore weight earlier-window results (W2-W4) more heavily through the cell-mean recovery ratio.

External validity. Inherited from Papers 4-7: synthetic EEHDA structural priors, 14-day window, 0.15 patch debt reduction, 0.02 EPSS random walk, 8% KEV prevalence. Real fleets with different rate-of-distributional-shift dynamics would produce different (η, H) optima; the qualitative finding, whether smoothing helps at high capacity, is more robust than the specific parameter choices.

X. Related Work

Lag-1 online calibration baseline. The online-calibration study [1] introduced the rolling-history grid-search recalibration procedure and reported the capacity-conditional hazard that motivates this paper. The present sweep inherits the online-calibration study's `fixed`/`lag1`/`offline` strategies and adds the two smoothing variants.

High-capacity collapse. our earlier capacity-decay study identified the high-capacity regime where HygienePrio-full collapses even with fixed weights. The present paper does not attempt to rescue the absolute floor (that question is taken up in the future-work section); instead it asks whether a smoothed online recalibration procedure can match or exceed the fixed-weight baseline at the collapse cells.

Exponential smoothing. The EWMA estimator dates back at least to Holt's seasonal-forecasting work [2]; in machine-learning settings it appears under the name "geometric averaging" as a low-variance bias-bearing substitute for sliding-window means. The use here, weighting a finite past history of grid-search calibration scores, is a direct adaptation; no novel estimator is claimed.

Multi-paper sequence. A connected line of our prior work, spanning the HygieneBench synthetic benchmark, the HygienePrio scorer [3], a temporal-stability study across rolling maintenance windows, a capacity-indexed decay analysis, and the online-calibration study [1], together defines the methodological substrate within which this paper sits. This paper extends the calibration-strategy dimension by one step (single-lag $\rightarrow$ finite-history weighted aggregation).

Policy mandates. CISA BOD 22-01 [6], 23-01 [7], NIST SP 800-40 Rev. 4 [8], and the 2026 DBIR [9] establish the operational scheduling regime under which any deployable recalibration procedure must operate.

XI. Conclusion

A pre-registered 2,250-row evaluation of multi-window-history calibration, EWMA-3 and trailing-mean-3, against the online-calibration study lag-1 baseline produces four hypothesis rejections and one clean finding.

The clean finding: multi-window smoothing does not rescue the high-capacity hazard documented in the online-calibration study. At $K = 200$, EWMA-3's cell-mean recovery ratio is -0.94, worse than `lag1`'s -0.66. At $K = 100$ where `lag1` recovered the offline ceiling ($\rho = +0.99$), EWMA-3 produces $\rho = -1.37$, a substantial reversal. At $K = 50$ where `lag1` matched the ceiling, EWMA-3 falls to half of it.

The mechanism: under fast per-window distributional shift, older calibration windows are systematically less representative of the target window than the most recent past window.

Averaging across multiple past windows reduces calibration-target variance but introduces bias from those older states, and the bias dominates. The result is monotone in history length: lag1 > trail3 ≈ EWMA-3 at every cell.

The sharpened recommendation: deployable online recalibration in the simulated bi-weekly regime should use the shortest possible history (lag-1 at most), and should be turned off entirely at high capacity. Future adaptive procedures (change-point detection, forgetting-factor Bayesian update, capacity-conditional gating) need to be developed and pre-registered to address the regimes the online-calibration study and this paper together identify as untouched by simple history-based smoothers.

Reproducibility. The runner, analysis, pre-registration protocol, and frozen 2,250-row results CSV are available at https://github.com/Harshavardhanmalla-Labs/research-papers/tree/main/paper8.

Future work. (i) Change-point-aware recalibration: detect distributional shifts and reset the history window. (ii) Adaptive η EWMA: tune the forgetting factor online to match the detected shift rate. (iii) Real fleet validation: re-establish the fixed-weight baseline on real exploitation-event data before any of this paper's findings about online procedures are taken to a deployment recommendation. (iv) Sensitivity sweep over η and trailing-history length H: the present paper held both at pre-registered values; a follow-up could map the (η, H) landscape and identify whether there is any choice for which the smoothing-helps regime can be reached.

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